

Education

Bachelor of Engineering with Second Class Honors

Bachelor's degree program

Major: Mechatronics

A multidisciplinary degree focused on the intersection of mechanical, electrical, and software engineering. My electives and final-year project emphasized automation, wireless communication, and data-driven control systems.

University Of Canterbury

January 2020 – November 2024

GPA: 6.50

Relevant Experience

Syos Aerospace

Mechatronics Intern

Nov 2024 – Mar 2025

Tauranga, NZ

Developed an embedded system to optimize data translation between messaging protocols for autonomous GPS navigation.

Used C++ for firmware development and Python to automate failure scenarios in HITL testing.

Conducted performance analysis through simulations and field tests, collecting data to model vehicle positioning accuracy. Delivered findings in a technical report to assist with system certification.

Gained practical experience assembling autonomous vehicles. Learned to document standard operating procedures.

Final Year Project (A) - Autonomous underwater Robot

Team of 4 | Client: Cawthron Institute

Full year 2024

University of Canterbury

Developed an underwater robot with motors, sensor payload and wireless recharging to support year-long deployments up to 30 meters deep.

Designed a wireless underwater charging system and implemented a Bluetooth communication interface using MicroPython on an ESP32.

Created custom electrical circuits to control robotic subsystems.

Cawthron (Marine Research Institute)

Engineering Intern

Nov 2023 – Feb 2024

Nelson, NZ

Created bespoke monitoring systems to support marine science research.

Developed a Python-based user interface for automated temperature sensor calibration.

Automated data pre-processing of large time-series datasets with using Pandas.

Presented technical work at internal research displays, strengthening communication and presentation skills.

Mechatronics Teaching Assistant

Electrical and Mechatronics Departments

November 2024 - Present

University of Canterbury

Supporting students in labs and tutorials across PCB design, embedded systems programming, and PLC systems.

Developed strong communication skills and the ability to explain complex topics.

UC Aerospace

Treasurer

2024

As treasurer I managed project spending and fulfilled our financial disclosure obligations as a registered charity. I volunteered for the role to develop better financial planning skills and support my friends. Our major project was the international rocketry competition (Spaceport America) held in the USA, where we won our category (30,000 ft commercial components).

Referees (Please email for written references)

Paul Barter	Senior Marine Scientist & Engineering Team Lead at Cawthron Email: Paul.Barter@cawthron.org.nz
Baris Danacioglu	Design Engineering Manager at Syos Aerospace Email: baris.danacioglu@syos-aerospace.com

Awards

2nd Place	Embedded Systems RC car competition	2024
Emerging Scientist Scholarship	Cawthron Marine Institute	2023
2nd Place	Solar Car Design Competition	2023
3rd Place	UC Motorsport RC Car Challenge	2022

Prior Work Experience

Teasme Tea	March-June 2022
<i>Cafe Front of House</i>	Christchurch, NZ

- Worked independently in a high-pressure environment to deliver excellent customer service
- Maintained a clean shop and kitchen.
- Can cure any ailment with the right tea

Firststeel (steel yard)	January-March 2022
<i>Yard Laborer</i>	Christchurch, NZ

- Advised customers purchasing steel and safely loaded trailers
- Assembled hydraulic trailers and steel-storage racks
- Gained experience working in hazardous and dynamic environments

Borderlandz Paintball	Sept 2019 - Dec 2020
<i>Referee</i>	Carterton, NZ

- Developed strong interpersonal skills to ensure a balance between having fun and following rules.
- Cleaned and maintained markers (guns), masks and fields
- Once decimated a stag-do team 2 v 12